Praxis — turning lecture material into adaptive mini-courses

COMP5241 Software Engineering · group project · week 9 · progress, division of work, system design

1. WHERE WE ARE

The mid-term prototype was six static screens. Since then the platform behind them has been built: a Next.js application, a Postgres database, a session with role guards, and server actions that make the instructor screens write to it. The screens themselves are unchanged — the prototype is the source of truth for the interface, and every screen in the app is generated from it by a converter rather than re-typed.

Layer	State	Evidence
Interface, six screens	complete, visually identical to the prototype	`prototype/` has one commit and no edits since
Application, Next.js 15 App Router	three surfaces wired to the database	course editor 55 of 57 controls, `/admin` 5 of 5, `/settings` 6 of 6
Data, Postgres on Neon + Prisma	live and seeded	6 courses · 20 modules · 22 activities · 24 learners
AI generation pipeline	not started	scheduled week 10
Deployment, automated tests	pipeline written, never executed	scheduled week 11

2. SYSTEM DESIGN

```
browser ──form POST──▶ server action ──▶ lib/store.ts ──▶ Prisma ──▶ Neon Postgres
                              │
                              └── revalidatePath ──▶ the same page re-renders
```

No hand-written API layer. Every mutation is a server action. The form posts, the action validates, the store writes, the page re-renders. There is no REST endpoint to keep in sync with the UI and no client-side fetch for course data, so a screen cannot drift out of step with the schema it writes to.

The prototype owns the interface. `tools/html_to_tsx.py` converts `prototype/*.html` into the React components under `components/ported/`. Behaviour is added around the markup, not inside it. This is why the app looks exactly like the artefact reviewed in week 7, and why a design change is a one-line edit in one file followed by a re-run, not a sweep through six screens.

Ten tables, one writer. `users`, `cohorts`, `courses`, `modules`, `activities`, `enrollments`, `events`, `course_versions`, `ai_runs`, `prompt_versions`. `lib/store.ts` is the only module that writes; pages read through it. The unique `(courseId, position)` constraint on modules is why reordering and deletion need renumbering, and that logic lives in one place because of it.

3. AGAINST THE BRIEF

Requirement	State
R1–R3 mini-courses, topic coverage, seven activity formats	interface complete; five of seven formats modelled; games and AI narration specified, not built
R4–R5 audience adaptation and personalisation	rules and registers present in the UI; execution needs the generation step
R6–R7 instructor editing and AI course generation	editing is live and writes to the database; generation week 10
R8 cohort import and engagement tracking	cohorts and instructor assignment live; Moodle/CSV import still a mock
R10 learner-defined paths	screen present, behaviour week 10
R11 iterative prototyping	round 1 wireframes → round 2 hi-fi prototype → this build
R12 each member contributes visibly	ownership table in `midterm-update.md` §7; issues and PRs per feature
R13 public repository	present; the platform branch is merged in week 10
R14 stakeholder and user interviews	scheduled this week
R15 AI Fluency reflection	`docs/ai-usage-log.md`
R16 deploy to a public platform	pipeline ready, production URL week 11

4. DIVISION OF WORK

Four vertical slices, one per member, each owning a feature from screen to data rather than a horizontal layer.

Workstream	Covers	Where it shows
Product & AI pipeline	framing, prompt registry, generation checks, evaluation	generation flow, `ai-usage-log.md`
Learning runtime	player, code lab, hint ladder, activity state	the learner screen
Instructor surface	editor, cohorts, analytics	editor, cohort table, analytics
Platform & experience	design system, auth, deployment, CI	design system, build and deploy pipeline

Shared and rotating: requirement interviews, prototype reviews, documentation, slides. Names and student IDs are on the week 3 group-formation form.

5. WHAT WE ARE NOT CLAIMING

- **Three surfaces write to the database, four do not.** The course editor, the cohort registry and settings are live. The studio form, the learner player, the path builder and the analytics page are still prototype markup. We will demo the first three and describe the rest as designed, not built.
- **There is no authentication.** The session is a role switcher for demonstration; it is not a login. Real accounts are out of scope for week 11.
- **No AI call has been made.** Every outline seen so far is fixture data. The generation step lands in week 10, and the interface will say which mode produced a course.

6. NEXT THREE WEEKS

Week	Work
9	learner runtime and hint ladder; interviews; merge the platform branch
10	generation pipeline against a real model; path builder; analytics queries
11	deploy, automated smoke tests, demo video, slides, report, user manuals

Evidence for every claim above is in the repository: `docs/architecture.md`, `docs/test-plan.md` §8 (what was actually run, with output), `docs/requirements.md` (traceability), `docs/midterm-update.md` §7 (ownership).